

CO3-AT1-Q1:

Name: D.Uday Kiran

Reg no: 192425161

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Course Name: DAA for Complexity Analysis

Quick Sort Worst Case Analysis

Problem Understanding

Quick Sort is a Divide and Conquer sorting algorithm. In the worst case, selecting the smallest or largest element as pivot repeatedly creates unbalanced partitions.

Analysis

For sorted datasets:

- Pivot selection becomes poor.
- One side contains $n-1$ elements.
- Recursion depth increases.

Therefore:

Worst Case Time Complexity = $O(n^2)$

Random datasets produce balanced partitions:

Average Complexity = $O(n \log n)$

Solution

Random pivot selection avoids repeated bad partitions and improves performance.

Techniques:

- Random pivot selection
- Median-of-three pivot selection
- Hybrid sorting methods

Conclusion

Quick Sort performs efficiently for random data but may degrade for already sorted datasets. Random pivot selection reduces the possibility of worst-case behavior.